

Korean Petrochemical Industry

Decarbonization Pathway Analysis

Marginal Abatement Cost Curve (MACC) Model

Report Generated: 2025-10-11 23:27

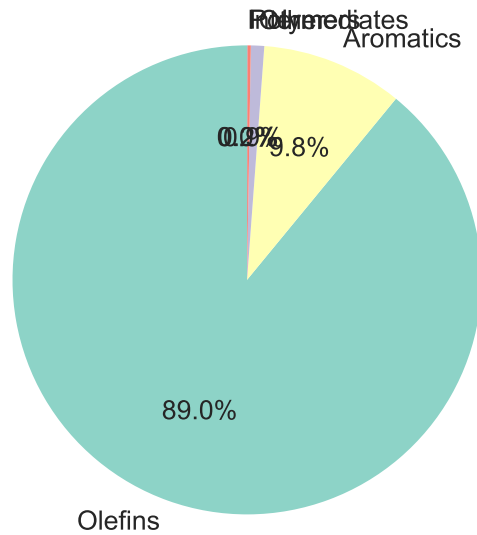
Model Version: 2.2

Key Model Parameters

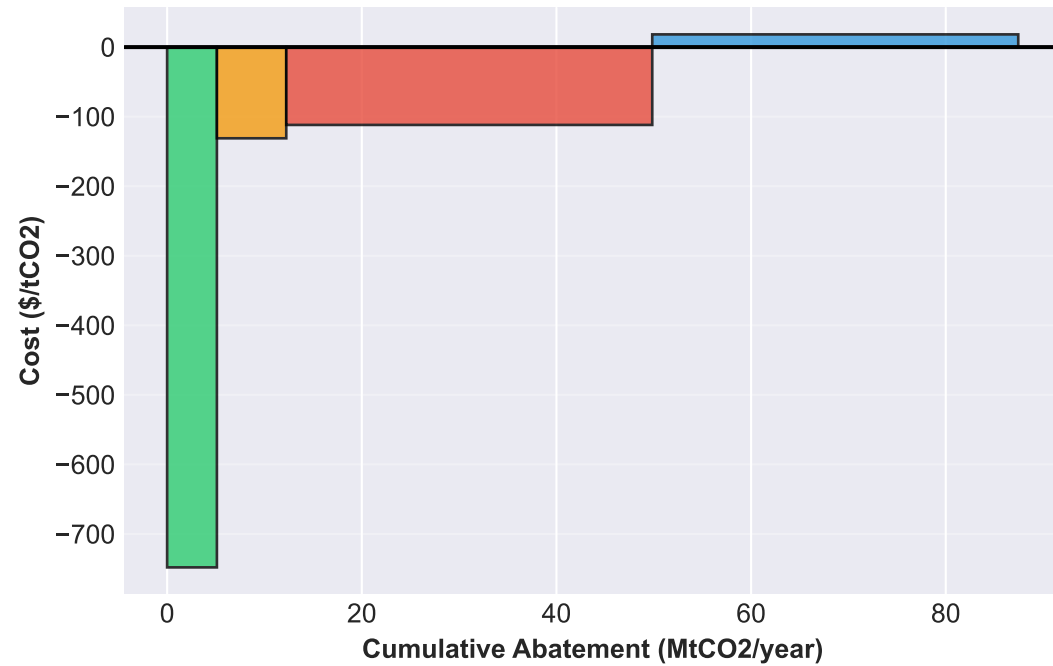
- Baseline Emissions (2025): 52.0 MtCO₂/year
 - Facilities Analyzed: 248
- Technologies: 4 (Heat Pump, RE PPA, NCC-H₂, NCC-Electricity)
 - Time Horizon: 2025-2050
- Maximum Abatement Potential: 49.8 MtCO₂ (95.9%)
 - Scenarios: Conservative, Moderate, Aggressive

Executive Summary

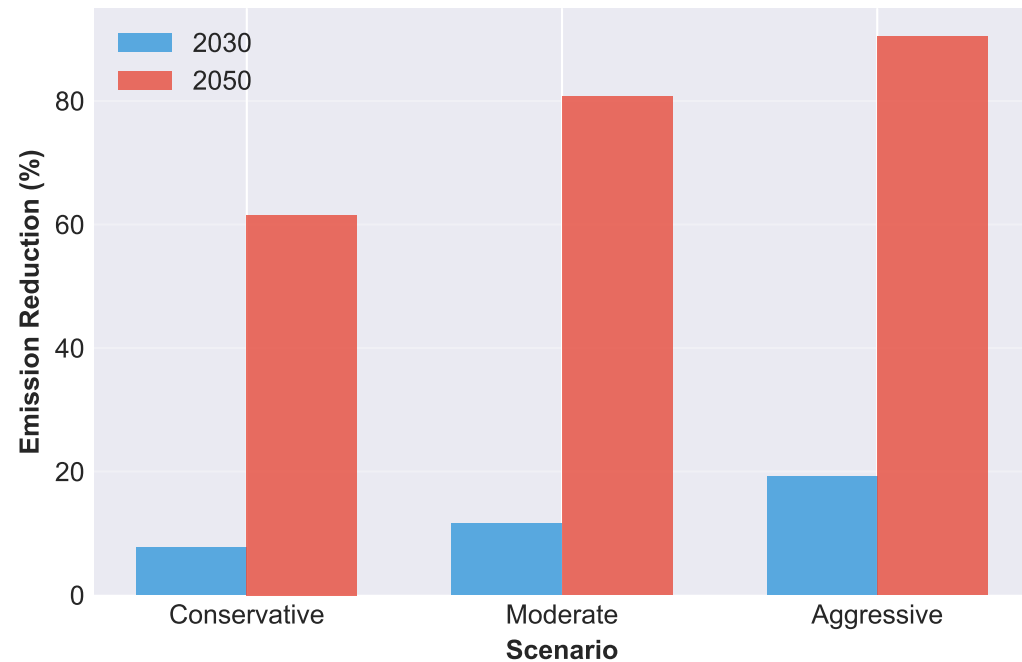
2025 Baseline by Product Group (52.0 MtCO2 total)



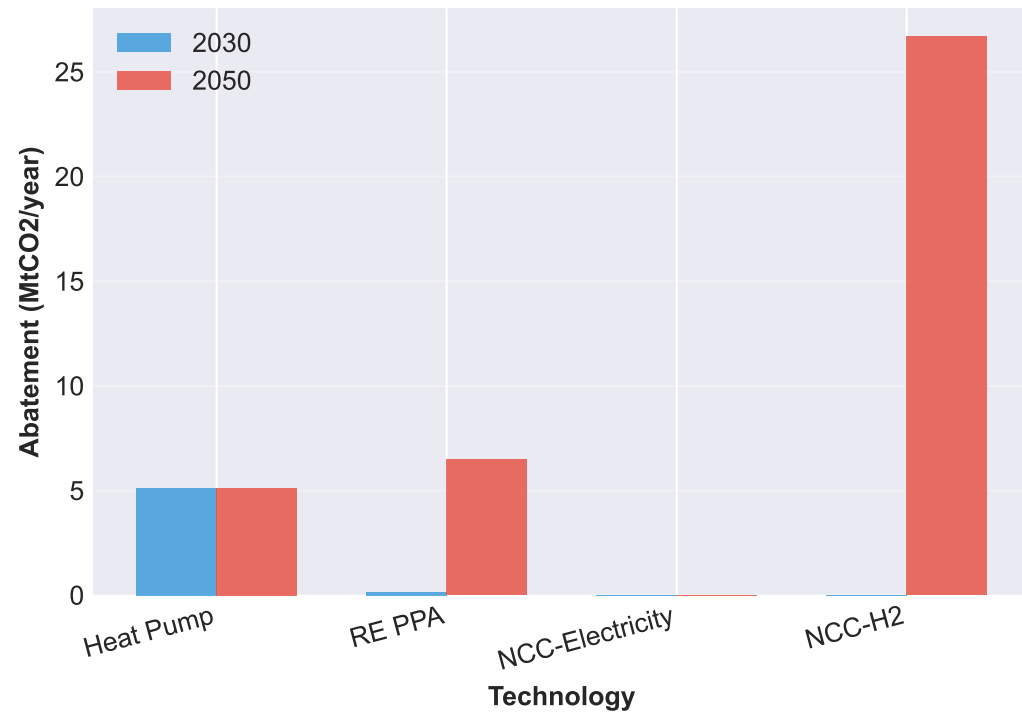
MACC Curve - 2030 (3 cost-negative technologies)



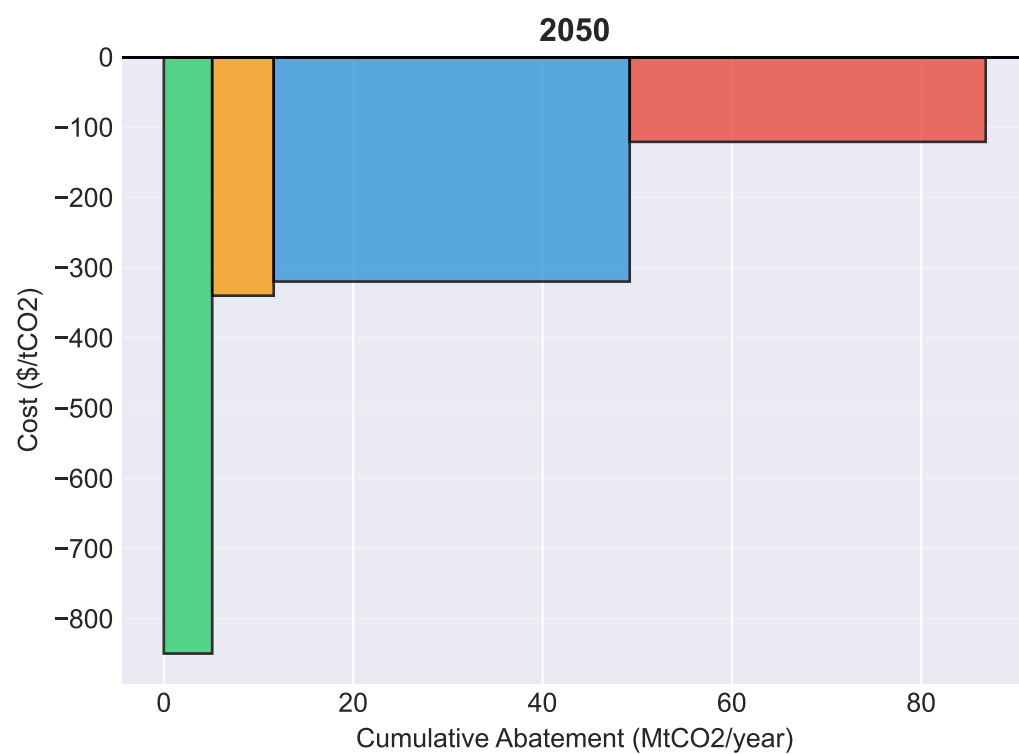
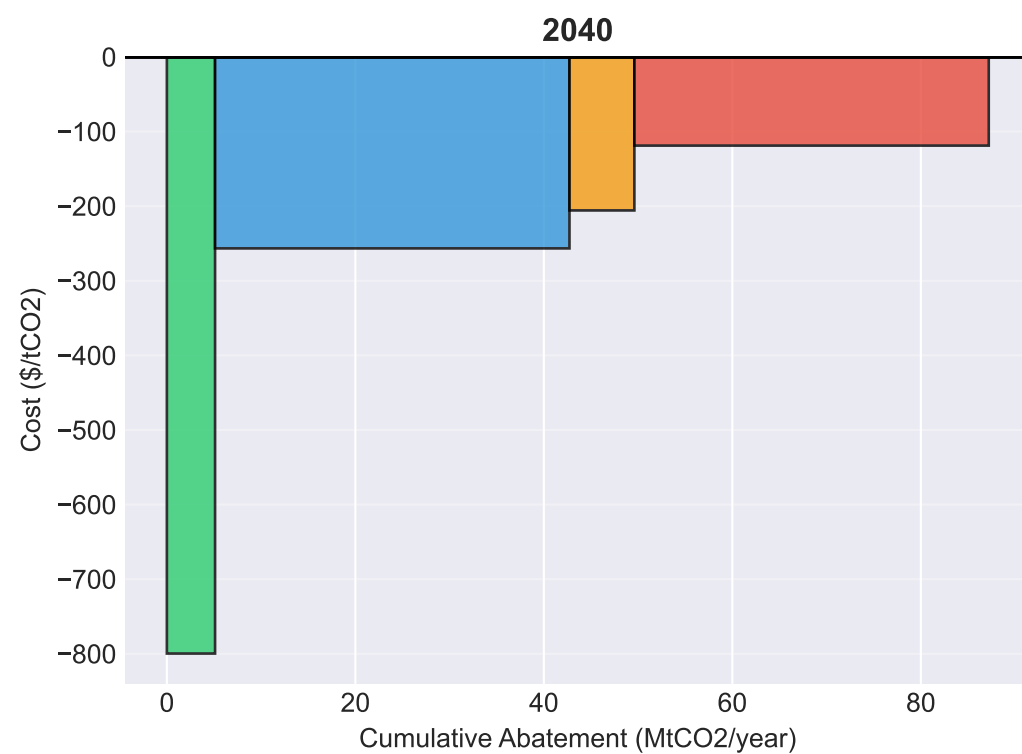
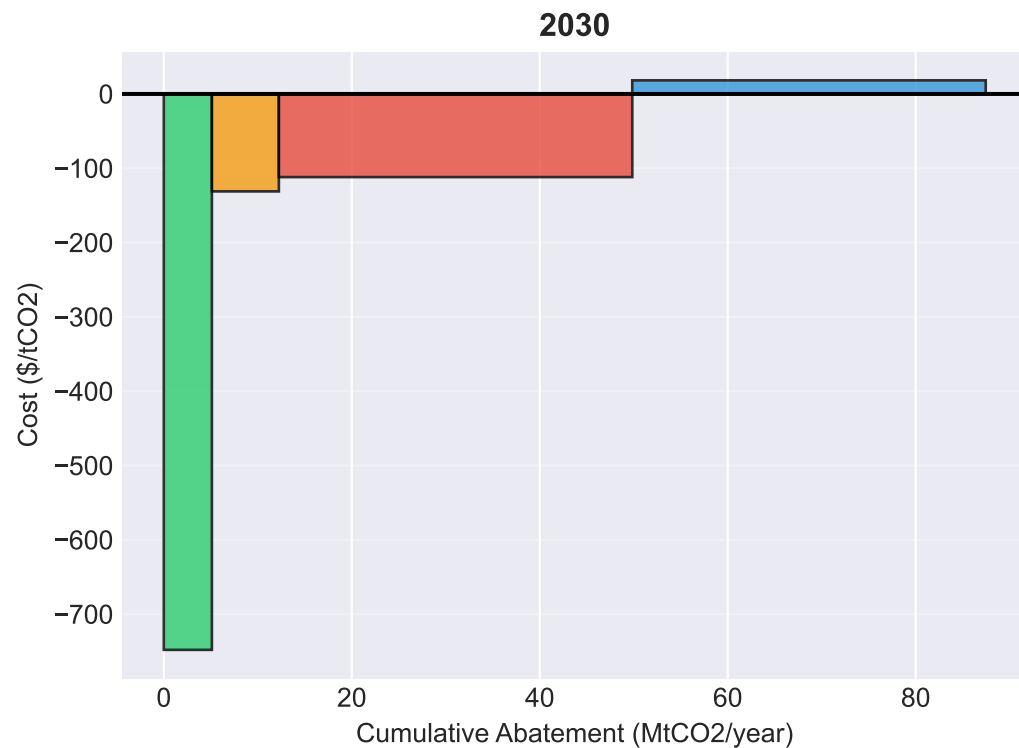
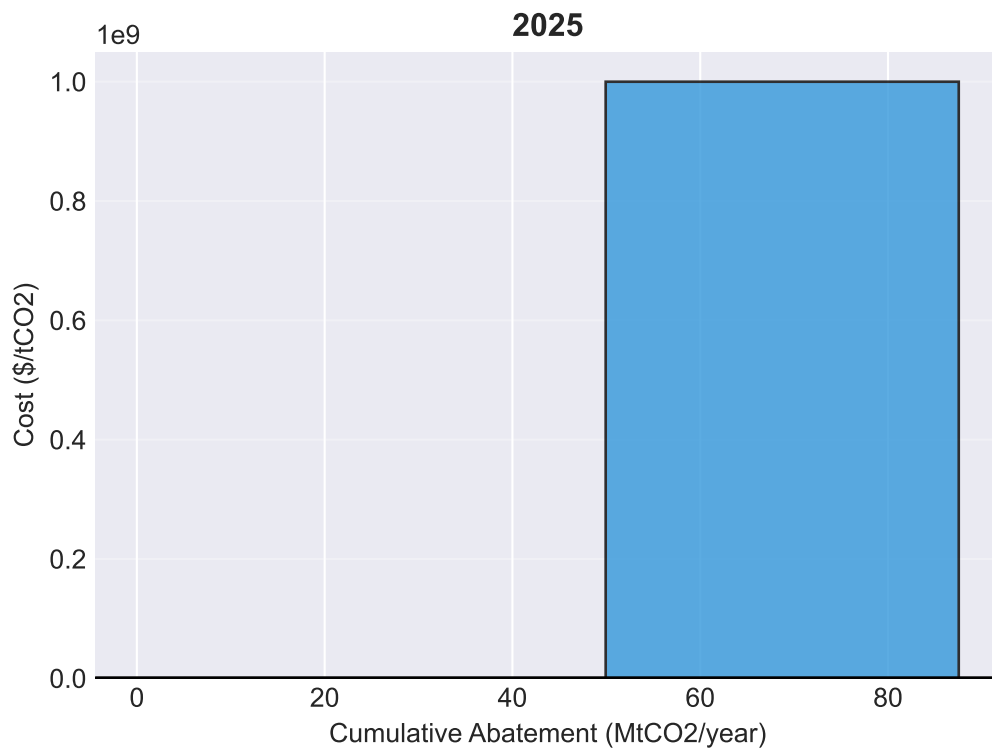
Emission Reduction by Scenario



Technology Deployment - Moderate Scenario

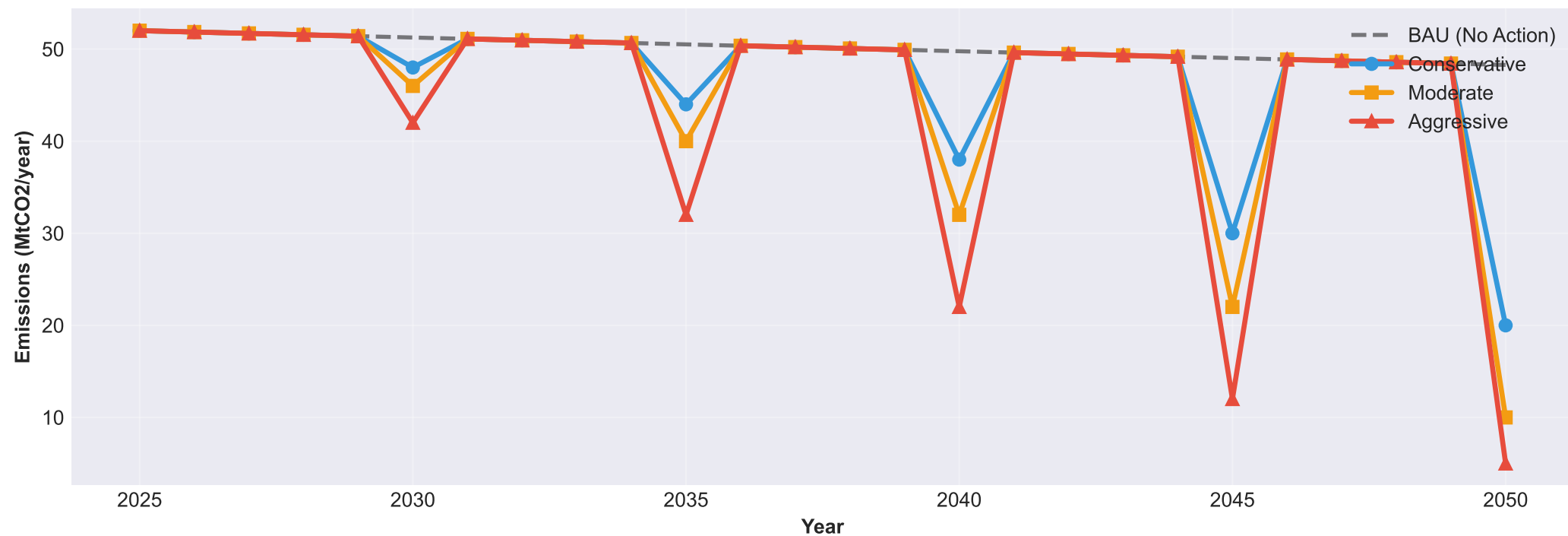


MACC Curve Evolution (2025-2050)



Emission Trajectories by Scenario

Annual Emissions



Technology Deployment - Moderate Scenario

